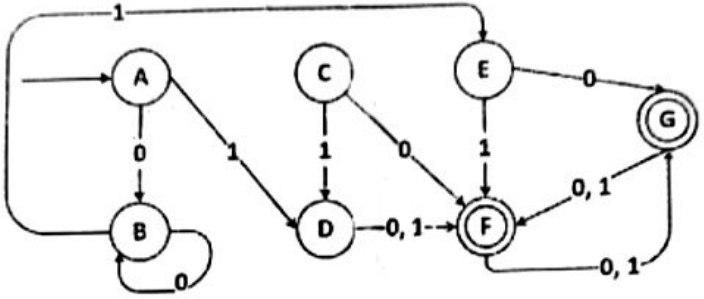
B. Tech. (Computer Engineering) Vth Semester
1st Sessional, AUTOMATA THEORY (CEN-502)

Time: 1 Hour **MM:15**

Note: Attempt all questions. **Date: 11/09/24**

Your Roll No : .....

Q No.	CO Mapped
1	1
2	1
3	2
4	2

Q. No	Question	Marks
1.	Design a minimum DFA which generates binary strings ($\sum$ = {0, 1}) in which there exist equal number of '001' & '100'. *Note: Strings like 11001100110 & 00100 are also acceptable in this DFA.* ***Verify your answer using suitable examples.***	5
2.	Minimize the following DFA using the Matrix Method showing all intermediate steps.  ***Verify your minimized answer using suitable examples.***	3
3.	Generate the Regular expression for the **Minimum DFA** obtained for Q2 given above.	3
4	State as well as apply Pumping Lemma, to comment on the Regularity/ Non Regularity of the following language which generate strings as follows : $L = \{ (ab)^{3n} \mid n > 0 \}$	4